

DEVAYAN MANDAL

Full Stack AI Native / Forward Deployed AI Engineer & Open Source Contributor - Open to Remote Work

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## PROFESSIONAL SUMMARY

Full Stack AI Native / Forward Deployed AI Engineer with 10 years of remote, self-directed production work. Built 30+ production-grade systems across autonomous multi-agent platforms, MCP integrations, deterministic RL evaluation, real-time observability, and high-throughput web applications. Merged 20 upstream pull requests across TypeScript, JavaScript, Python, and Rust in Recharts, pnpm, node-postgres, SQLMesh, Biome, Better Auth, thirdweb, Anza Kit, and Rspress. Specializes in explicit state machines, bounded autonomy, verifiable AI systems, hybrid RAG and memory architectures, streaming infrastructure, and production-grade developer tooling, with end-to-end ownership from architecture through deployment, observability, testing, and upstream delivery.

## TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Rust, SQL, R, Bash

AI & Agents: LangGraph, LangChain, LangServe, LangSmith, Model Context Protocol (MCP), multi-agent orchestration, supervisor routing, Reflexion loops, human-in-the-loop gates, hybrid RAG, RLVR environments, deterministic evaluation, prompt engineering, Ollama, Gemini, OpenAI-compatible providers

Frontend: React 19, Next.js 16, TanStack Start/Router/Query, Tailwind v4, Vite, Framer Motion, Expo, Three.js, React Three Fiber, HTML5 Canvas, SVG

Backend & Data: FastAPI, Node.js, Drizzle ORM, Prisma, SQLAlchemy, PostgreSQL (pgvector), Neon, Supabase, SQLite, Pydantic, Zod, Stripe

Realtime & Observability: SSE, WebSockets, Socket.io, replayable traces,

OpenTelemetry-shaped tracing, LangSmith evaluation pipelines

Cloud & DevOps: Vercel, AWS, Microsoft Azure, GitHub Actions CI/CD, serverless, scheduled pipelines, Vitest, pytest, Playwright, Biome

Web3 & Blockchain: Solidity, OpenZeppelin, Foundry, viem, wagmi, ethers.js, Safe Protocol Kit, RainbowKit, Reown AppKit, Base L2, blockchain and wallet infrastructure

Practices: Six Sigma DMAIC, bounded autonomy, retrieval-bounded generation, AST-sandboxed grading, cost-aware routing, threat-model documentation, Git, Changesets

## PROFESSIONAL EXPERIENCE

### LATTICE AI LABS

Full Stack AI Native / Forward Deployed AI Engineer | January 2016 - Present | Remote

- Architected and shipped 30+ production-grade systems across a broad portfolio of public engineering repositories spanning autonomous agents, RL evaluation environments, real-time observability, and high-throughput web applications.

- Built AEGIS, an enterprise AI ops cortex where a LangGraph supervisor routes 6 specialist agents across 7 agentic loops, with hybrid RAG, human-in-the-loop gates, LangSmith evaluation gates, and a live SSE dashboard. Live: [aegis-agent-api.vercel.app/ui](https://aegis-agent-api.vercel.app/ui)

- Built Synthesis, an autonomous research platform running cyclic Reflexion workflows (Plan, Research, Synthesize, Critique, Finalize) on a StateGraph executor with HITL approval gates, and AAROP, an inspectable 6-phase reasoning engine with cost-aware model routing, circuit breakers, RAG memory, replayable traces, and 99% Python test coverage. Live: [synthesis-gold.vercel.app](https://synthesis-gold.vercel.app) | [aarop.vercel.app](https://aarop.vercel.app)

- Built Veritas, a LangGraph research agent with SSE streaming, grounded demo mode, external retrieval when configured, and explicit separation between deterministic demo execution and live model/retrieval execution. Live: [veritas-engine-woad.vercel.app](https://veritas-engine-woad.vercel.app)

- Authored 3 official RLVR environments on the Prime Intellect Environments Hub (calendar-math, meeting-slot, unit-fix) using deterministic UTC sweep-line solvers and AST-sandboxed execution graders, removing LLM-as-judge variance.

- Implemented MCP servers and clients across Stdio and StreamableHTTP/SSE transports with client-side sampling and roots-based security boundaries; shipped HoldSlot, a WebMCP implementation for the OpenAI WebMCP Challenge 2026.

- Engineered Cadence, a framework-free healthcare agent with 5-layer clinical memory (session, episodic, semantic, knowledge, insights) using Ollama and SQLite, producing consent-scoped clinician briefs. Live: [cadence-healthcare.vercel.app](https://cadence-healthcare.vercel.app)

- Shipped Jobrow, a live US tech job board indexing 5,000+ still-open roles across employer ATS boards (Greenhouse, Ashby, Lever) on TanStack Start, React 19, and Neon Postgres, with a public JSON API, Expo iOS/Android client, and GitHub Actions crawl pipeline running 4x daily. Live: [jobrow.vercel.app](https://jobrow.vercel.app)
- Shipped Lattice, a live Web3 job platform for blockchain, crypto, and DeFi roles with same-day close detection from employer ATS boards. Live: [lattice-devtechedge1.vercel.app](https://lattice-devtechedge1.vercel.app)
- Built a real-time chaos engineering platform with self-healing microservices, animated SVG topology, automated fault injection, and sub-second WebSocket telemetry, plus CollabSpace multiplayer whiteboards on Supabase Realtime; wired replayable OTel-shaped traces and LangSmith evaluation pipelines through the agent systems. Live: [chaos-simulation.vercel.app](https://chaos-simulation.vercel.app)
- Delivered Pulsar, a Base L2 GPU marketplace with Foundry smart contracts; RegTrace-AI, a MiCA/VARA compliance copilot with source-traced citations and hallucination flags; Smart Waitlist with Stripe-backed referral flows; and AstraMart.
- Operate a serverless daily AI news agent on GitHub Actions that pulls six public feeds, dedupes against in-repo memory, summarizes with Gemini, and delivers one Telegram executive brief at zero infrastructure cost.
- Work AI-native across the full stack using frontier coding and reasoning models through dedicated development harnesses, with emphasis on rapid iteration, test-driven validation, repository-specific contribution policies, and reproducible upstream delivery.

## SPECTRAL QUANTITATIVE TRADING

Quantitative Crypto Trader & Derivatives Strategist | September 2016 - Present | Remote

- Manage diversified proprietary capital across tier-1 exchanges including Binance, Coinbase, Bitget, and MEXC, and on-chain protocols including Hyperliquid, Uniswap v3, Aave, and GMX.
- Built and ran automated grid, DCA, and Martingale bots on Pionex and Binance to capture spread and volatility in ranging markets.
- Executed delta-neutral basis and funding-rate arbitrage between spot and perpetuals, plus concentrated-liquidity and lending-loop strategies on-chain.
- Applied TWAP/VWAP execution on illiquid pairs, with VaR limits, drawdown caps, and Sharpe-ratio optimization governing position sizing and portfolio risk.

## OPEN SOURCE CONTRIBUTIONS

Merged upstream:

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- PyO3/maturin #3302 (TypeScript) - `find_all_windows()` now lossy-decodes py --list-paths and conda info -e output instead of unwrapping `str::from_utf8`, so a Windows build with localized (non-UTF-8) interpreter output skips the affected line rather than panicking and aborting the build; added a unit test covering the non-UTF-8 decode path. Merged Sep 2026. [github.com/PyO3/maturin/pull/3302](https://github.com/PyO3/maturin/pull/3302)
- stellar/stellar-docs #2859 (MDX) - added the --enable-core-manual-close flag to the Local section of the Network Modes page, so the parameter list no longer omits a flag the quickstart container ships, with a cross-link to the Manual close mode section noting that only the local configuration sets `NODE_IS_VALIDATOR`. Merged Sep 2026. [github.com/stellar/stellar-docs/pull/2859](https://github.com/stellar/stellar-docs/pull/2859)
- stellar/stellar-docs #2850 (MDX) - documented Quickstart's undocumented '--enable-core-manual-close' flag in the advanced usage docs, covering the 'MANUAL\_CLOSE' setting it writes into the generated core config, the 'manualclose' endpoint that advances one ledger per call, and loopback-bound startup examples for macOS, Linux, and Windows. Merged Sep 2026. [github.com/stellar/stellar-docs/pull/2850](https://github.com/stellar/stellar-docs/pull/2850)
- ssf0409/tracelens #140 (Python) - markdown table cells in the report generator are now escaped with `html.escape` in addition to pipe and newline handling. Task ids or gate values containing pipes, line breaks, or HTML metacharacters no longer break the per-task and baseline-gate tables piped into `$GITHUB_STEP_SUMMARY`, and no cell can open a raw HTML element. Merged Sep 2026. [github.com/ssf0409/tracelens/pull/140](https://github.com/ssf0409/tracelens/pull/140)
- stellar/stellar-docs #2849 (MDX) - reworked the address conversion example to propagate the fallible 'Result<Address, ConversionError>' instead of unwrapping malformed XDR, and documented that untrusted XDR must be handled rather than panicking. Merged Sep 2026. [github.com/stellar/stellar-docs/pull/2849](https://github.com/stellar/stellar-docs/pull/2849)
- stellar/stellar-docs #2853 (MDX) - reconciled memo framing in the pooled accounts guide so memos read as the legacy mechanism still in active use and muxed accounts as preferred going forward. Merged Sep 2026. [github.com/stellar/stellar-docs/pull/2853](https://github.com/stellar/stellar-docs/pull/2853)
- stellar/stellar-docs #2851 (MDX) - qualified the Freighter HTTPS requirement, noting that

'http://localhost' is already a Potentially Trustworthy origin under W3C Secure Contexts so loopback needs no TLS. Merged Sep 2026. [github.com/stellar/stellar-docs/pull/2851](https://github.com/stellar/stellar-docs/pull/2851)

- pytest-dev/pytest-env #262 (Python) - documented that pytest.toml and .pytest.toml accept the native [pytest] env table, not only the plugin-specific [pytest\_env] section. Merged Sep 2026. [github.com/pytest-dev/pytest-env/pull/262](https://github.com/pytest-dev/pytest-env/pull/262)
- pnpm/pnpm #14863 (Rust/TypeScript) - fixed startup crashes on FreeBSD and other non-Windows Unix-like platforms by making default\_store\_dir use the Unix fallback path; added platform-specific regression coverage. Merged Sep 2026. [github.com/pnpm/pnpm/pull/14863](https://github.com/pnpm/pnpm/pull/14863)
- web-infra-dev/rspress #3678 (TypeScript) - fixed a Search initialization race by awaiting FlexSearch addAsync indexing before LocalProvider.init() resolves, preventing early searches from incorrectly returning no results on larger sites; focused regression-oriented fix. Merged Sep 2026. [github.com/web-infra-dev/rspress/pull/3678](https://github.com/web-infra-dev/rspress/pull/3678)
- recharts/recharts #7805 (TypeScript) - removed tabIndex={-1} from z-index portal SVG <g> layers so empty groups are not pointer-focusable and WebKit does not draw geometry-traced focus rings; added regression coverage asserting no layer carries a tabIndex attribute. Merged Sep 2026. [github.com/recharts/recharts/pull/7805](https://github.com/recharts/recharts/pull/7805)
- brian/node-postgres #3772 (JavaScript) - stopped Connection.sync() from incorrectly setting the internal \_ending flag, preventing false suppression of subsequent socket errors; added unit coverage for Sync vs disconnect semantics. Merged Sep 2026. [github.com/brian/node-postgres/pull/3772](https://github.com/brian/node-postgres/pull/3772)
- pnpm/pnpm #14756 (Rust/TypeScript) - preserved existing dependency range operators and npm:/jsr: protocol prefixes during pnpm update, including prerelease and exact-version handling. Merged Sep 2026. [github.com/pnpm/pnpm/pull/14756](https://github.com/pnpm/pnpm/pull/14756)
- pnpm/pnpm #14754 (Rust) - fixed non-recursive pattern runs with --no-bail so matching scripts continue executing and failures are aggregated correctly; added concurrent and sequential regression tests. Merged Sep 2026. [github.com/pnpm/pnpm/pull/14754](https://github.com/pnpm/pnpm/pull/14754)
- pnpm/pnpm #14753 (Rust/TypeScript) - fixed lockfile:false being ignored when automatic package-manager switching downloaded the configured engine; added unit and filesystem-backed regression coverage across Rust and TypeScript paths. Merged Sep 2026. [github.com/pnpm/pnpm/pull/14753](https://github.com/pnpm/pnpm/pull/14753)
- SQLMesh/sqlmesh #6040 (Python) - moved ModelTest.create\_test() off concurrent worker threads to eliminate a time\_machine/ttl\_cache race in sqlmesh test; verified the focused race regression repeatedly. Merged Sep 2026. [github.com/SQLMesh/sqlmesh/pull/6040](https://github.com/SQLMesh/sqlmesh/pull/6040)
- anza-xyz/kit #2032 (TypeScript) - restored number predicates in the getPatternMatchCodec advanced guide to match current codec typing. Merged Sep 2026. [github.com/anza-xyz/kit/pull/2032](https://github.com/anza-xyz/kit/pull/2032)
- better-auth/better-auth #11208 (TypeScript) - added regression coverage locking the /phone-number/verify OpenAPI requestBody after a Zod .and(z.record()) intersection; tests only, no production generator changes. Merged Sep 2026. [github.com/better-auth/better-auth/pull/11208](https://github.com/better-auth/better-auth/pull/11208)
- biomejs/biome #11667 (Rust) - ported the unicorn better-dom-traversing rule into Biome's nursery linter; +1,972 lines across 17 files and 7 commits, released to 13 packages via changeset. Merged Sep 2026. [github.com/biomejs/biome/pull/11667](https://github.com/biomejs/biome/pull/11667)
- thirdweb-dev/js #8938 (TypeScript) - fixed useTokenQuery error handling so 401/429/timeout failures rethrow instead of rendering a false "Token Not Supported" state in CheckoutWidget; added tests and a patch changeset. Merged Sep 2026. [github.com/thirdweb-dev/js/pull/8938](https://github.com/thirdweb-dev/js/pull/8938)

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Active upstream engineering spans AI infrastructure, developer tooling, package management, frameworks, accessibility, concurrency, portability, security, SDKs, wallets, and blockchain infrastructure. Canonical ledger: [github.com/devtechedge/oss-contributions](https://github.com/devtechedge/oss-contributions)

## CERTIFICATIONS

- Model Context Protocol: Advanced Topics - Anthropic, 2026
- AI Fluency for Builders - Anthropic (with CodePath), 2026
- Agents and Workflows - OpenAI Academy, 2026
- Deep Agents with LangGraph - LangChain Academy, 2026
- Ambient Agents with LangGraph - LangChain Academy, 2026
- Microsoft Certified: Azure Fundamentals (AZ-900) - Microsoft, 2020
- AWS Fundamentals Specialization - Amazon Web Services (Coursera), 2020
- Data Science Specialization - Johns Hopkins University (Coursera), 2020
- Six Sigma Green Belt - Kennesaw State University (Coursera), 2020

## EDUCATION

### UNIVERSITY OF MUMBAI

Bachelor of Engineering (B.E.), Chemical Engineering (First Class) | [Graduation Year]  
Quantitative core in process dynamics, transport phenomena, mathematical modeling, and optimization; transitioned into software engineering through computational and systems-oriented work.